



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**INTELLIGENT PNEUMONIA DETECTION AND DIAGNOSIS  
FROM CHEST X-RAY IMAGES USING DEEP LEARNING**

**A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements  
for the Course of*

**MLA0408-DEEP LEARNING**

*to the award of the degree of*

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE AND ENGINEERING  
(AI&ML)**

**Submitted by**

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**Saveetha Institute of Medical and Technical Sciences**

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**SEPTEMBER 2026**

# **SIMATS ENGINEERING**

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## **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “**Intelligent Pneumonia Detection and Diagnosis from Chest X-ray Images using Deep Learning**” has been carried out by **D.BinduPriya(192425162),K.Gowthami(192525428),M.Sri Lakshmi(192525404)** under the supervision of **DR.D Sudhagar** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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# **SIMATS ENGINEERING**

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## **DECLARATION**

**We, D. Bindu Priya(192425162), K. Gowthami(192525428), M. Sri Lakshmi(192525404) and B. Jahnavi(192325043) of the Btech.Computer Science and Engineering SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Intelligent Pneumonia Detection and Diagnosis from Chest X-ray Images using Deep Learning” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.**

**Place: Chennai**

**Date:**

**Signature of the Students with Names**

**K. Gowthami (192525428)**

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# SIMATS ENGINEERING

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## Individual Contribution Statement

| Student / Register No.        | Specific Responsibility                            | Design & Development Contribution                                                                                                 | Testing & Analysis                                                             | Report Contribution                                    |
|-------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------|
| D. Bindu Priya<br>(192425162) | Module 1 – Data Collection and Image Preprocessing | Dataset collection, organization, image quality checking, resizing, normalization, augmentation and dataset preparation pipeline. | Preprocessing validation, class-balance inspection and image-quality analysis. | Chapter 1, Chapter 2 and Module 1 documentation.       |
| K. Gowthami<br>(192525428)    | Module 2 – Deep Learning Model Development         | CNN / transfer-learning model selection, training pipeline, hyperparameter configuration and model development.                   | Training/validation analysis, loss and accuracy monitoring, model comparison.  | Chapter 3 and model-development portions of Chapter 4. |
| M. Sri Lakshmi<br>(192525404) | Module 3 – Explainable AI and Prediction           | Prediction pipeline, probability output, explainability method such as Grad-CAM, result visualization and user-facing prediction. | Confusion matrix, class-wise metrics and interpretation of model decisions.    | Chapter 4, Chapter 5 and explainability documentation. |

The three modules are designed as connected stages of one system: reliable data preparation supports model development, while the trained model is followed by explainable prediction so that the output can be interpreted rather than treated as an unexplained classification.

# ABSTRACT

Pneumonia is a serious respiratory infection in which early identification can support timely clinical assessment and treatment. Chest X-ray imaging is widely used as part of the diagnostic process, but interpretation can be challenging because radiographic patterns may overlap with other pulmonary conditions and because workload can vary across healthcare settings. This project proposes an intelligent computer-aided system for pneumonia detection and diagnosis support from chest X-ray images using deep learning.

The project is organized into three modules. Module 1, handled by D. Bindu Priya, focuses on data collection and image preprocessing. The module establishes a structured image dataset, performs quality inspection, removes unsuitable samples where required, standardizes image dimensions and intensity values, and applies suitable augmentation to improve model generalization. Module 2, handled by K. Gowthami, develops the deep learning model using a convolutional neural network or transfer-learning architecture. The model is trained to distinguish normal and pneumonia-related chest X-ray patterns and is evaluated using standard classification measures.

Module 3, handled by M. Sri Lakshmi, focuses on explainable AI and prediction. The trained model produces a predicted class and confidence/probability information, while an explainability technique such as Gradient-weighted Class Activation Mapping (Grad-CAM) can highlight image regions that contributed strongly to the prediction. This makes the output easier to inspect and supports responsible use of the system as a decision-support tool.

The overall workflow consists of data collection, preprocessing, augmentation, model training, validation, testing, prediction and explanation. The report emphasizes that model performance must be established using held-out data and appropriate metrics, and that an AI prediction should not be treated as a standalone clinical diagnosis. Future improvements include larger and more diverse datasets, external validation, calibration, multi-class disease classification, improved explainability, and integration into clinical workflows.

**Keywords:** Pneumonia Detection; Chest X-Ray; Deep Learning; Convolutional Neural Network; Transfer Learning; Image Preprocessing; Explainable AI; Grad-CAM; Medical Image Classification.

## TABLE OF CONTENTS

| Sl. No. | Title                                                                  | Page |
|---------|------------------------------------------------------------------------|------|
| i       | CERTIFICATE FROM THE SUPERVISOR                                        | ii   |
| ii      | DECLARATION BY THE CANDIDATE                                           | iii  |
| iii     | INDIVIDUAL CONTRIBUTION STATEMENT                                      | iv   |
| iv      | ABSTRACT                                                               | v    |
| v       | LIST OF FIGURES                                                        | vi   |
| vi      | LIST OF TABLES                                                         | vii  |
| vii     | LIST OF ABBREVIATIONS                                                  | viii |
| 1       | CHAPTER 1: Introduction and Engineering Problem                        | 1    |
| 2       | CHAPTER 2: Literature Review and Existing Solutions                    | 5    |
| 3       | CHAPTER 3: Engineering Design, Trade-offs, Ethics, Safety and Security | 9    |
| 4       | CHAPTER 4: Implementation, Testing, Results and Project Management     | 14   |
| 5       | CHAPTER 5: Conclusion, Future Work and Reflection                      | 20   |
|         | References                                                             | 23   |
|         | Appendices                                                             | 25   |

## LIST OF FIGURES

| Figure No. | Figure Name                                                     | Page No. |
|------------|-----------------------------------------------------------------|----------|
| 1.1        | System block diagram of the proposed pneumonia detection system | 3        |
| 1.2        | Overall module workflow                                         | 4        |
| 3.1        | Overall system architecture / data-flow diagram                 | 10       |
| 3.2        | Preprocessing and augmentation pipeline                         | 11       |
| 3.3        | CNN / transfer-learning model workflow                          | 12       |
| 4.1        | Sample preprocessing stages                                     | 15       |
| 4.2        | Training and validation accuracy/loss curves                    | 17       |
| 4.3        | Confusion matrix of the trained classifier                      | 18       |
| 4.4        | Grad-CAM explanation example                                    | 19       |
| 4.5        | Project timeline / Gantt chart                                  | 21       |

## LIST OF TABLES

| Table No. | Table Name                                                      | Page No. |
|-----------|-----------------------------------------------------------------|----------|
| 2.1       | Comparative analysis of existing pneumonia-detection approaches | 6        |
| 3.1       | Functional and non-functional requirements                      | 9        |
| 3.2       | Design specifications                                           | 10       |
| 3.3       | Alternative-solution evaluation matrix                          | 11       |
| 3.4       | Trade-off analysis                                              | 12       |
| 3.5       | Risk register                                                   | 13       |
| 4.1       | Dataset and preprocessing plan                                  | 15       |
| 4.2       | Test plan and verification criteria                             | 16       |
| 4.3       | Model evaluation metrics                                        | 18       |
| 4.4       | Achievement of project objectives                               | 20       |
| 4.5       | Roles and contribution summary                                  | 21       |



## LIST OF ABBREVIATIONS / SYMBOLS

| Symbol   | Description                                    | Unit |
|----------|------------------------------------------------|------|
| AI       | Artificial Intelligence                        | —    |
| CNN      | Convolutional Neural Network                   | —    |
| X-ray    | Radiographic imaging using X-rays              | —    |
| ROI      | Region of Interest                             | —    |
| DICOM    | Digital Imaging and Communications in Medicine | —    |
| GPU      | Graphics Processing Unit                       | —    |
| Grad-CAM | Gradient-weighted Class Activation Mapping     | —    |
| ReLU     | Rectified Linear Unit                          | —    |
| ROC      | Receiver Operating Characteristic              | —    |
| AUC      | Area Under the Curve                           | —    |
| TP       | True Positive                                  | —    |
| TN       | True Negative                                  | —    |
| FP       | False Positive                                 | —    |
| FN       | False Negative                                 | —    |

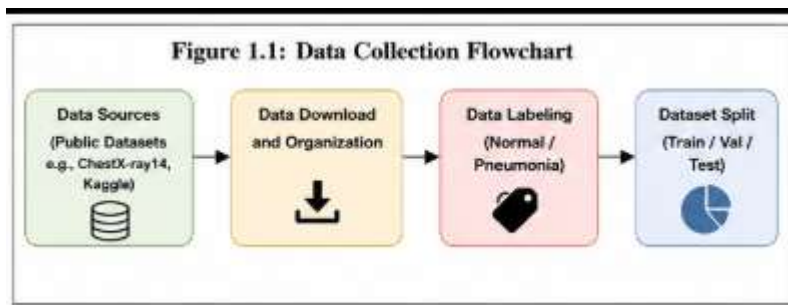
# CHAPTER 1

## INTRODUCTION AND ENGINEERING PROBLEM

### 1.1 Background

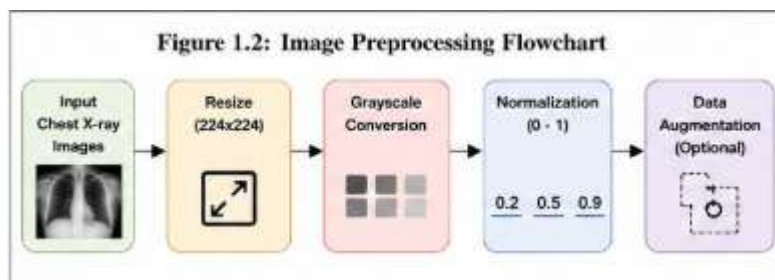
Pneumonia is an infection affecting the lungs and can produce radiographic abnormalities that require careful interpretation. Chest X-ray is an important imaging modality because it is comparatively accessible, fast and useful for assessing lung abnormalities. However, interpretation can be affected by image quality, overlapping patterns, clinical workload and differences in reader experience.

Recent progress in artificial intelligence has enabled deep learning systems to learn discriminative image features directly from medical images. Convolutional neural networks are particularly suitable for image classification because convolutional filters can learn hierarchical visual representations. Transfer learning can further reduce the amount of training required by starting from a model that has learned general visual features.



### 1.2 Need for the Project

The need for this project arises from the opportunity to provide a consistent computer-aided screening mechanism for chest X-ray images. Such a system can support prioritization and second-level review, particularly where large numbers of images need to be examined. The goal is not to replace a radiologist or physician but to provide an additional computational signal.



### 1.3 Problem Statement

The engineering problem is to design a deep-learning-based image analysis pipeline that can receive a chest X-ray image, perform reliable preprocessing, identify whether the image contains patterns associated with pneumonia, and present a prediction together with an interpretable visual explanation. The system must handle variation in image size, intensity, acquisition conditions and patient anatomy while controlling false predictions and avoiding misleading confidence.

## **1.4 Project Aim**

The aim is to develop an intelligent, modular and explainable pneumonia detection system using chest X-ray images, with separate modules for data preparation, deep learning model development, and explainable prediction.

## **1.5 Project Objectives**

- Collect and organize suitable chest X-ray images for model development.
- Preprocess and augment the images to produce a consistent training input.
- Develop and train a CNN or transfer-learning model for pneumonia classification.
- Evaluate the trained model using accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC where appropriate.
- Generate predictions with class probabilities and explain model attention using an explainability method such as Grad-CAM.
- Document limitations, ethical considerations and future improvements for responsible medical AI.

1.6 Scope

The project scope covers chest X-ray image classification for pneumonia-related screening. It includes dataset preparation, preprocessing, augmentation, model training, evaluation, prediction and explainability. The system is intended as a research and academic prototype that demonstrates the complete AI workflow.

The scope does not include autonomous clinical diagnosis, prescription of treatment, direct control of medical equipment, or replacement of qualified medical interpretation. External clinical validation and prospective hospital deployment are outside the present project scope.

1.7 Expected Engineering Outcome

The expected outcome is a reproducible pipeline in which an input chest X-ray is standardized, passed through a trained deep learning classifier and returned with a prediction and interpretable visualization. The design should also produce measurable evaluation results and a clear record of the data-processing steps.

1.8 System Workflow

| Stage             | Description                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------|
| Data Collection   | Gather and organize normal and pneumonia-labelled chest X-ray images from an appropriate dataset. |
| Preprocessing     | Resize, normalize, quality-check and standardize images.                                          |
| Augmentation      | Apply medically reasonable transformations to improve robustness.                                 |
| Model Development | Train CNN / transfer-learning classifier.                                                         |
| Evaluation        | Measure performance on validation and held-out test data.                                         |
| Explainability    | Generate heatmaps or region-based explanations for predictions.                                   |
| Prediction        | Display class and probability/confidence information.                                             |

### 1.9 Module Distribution

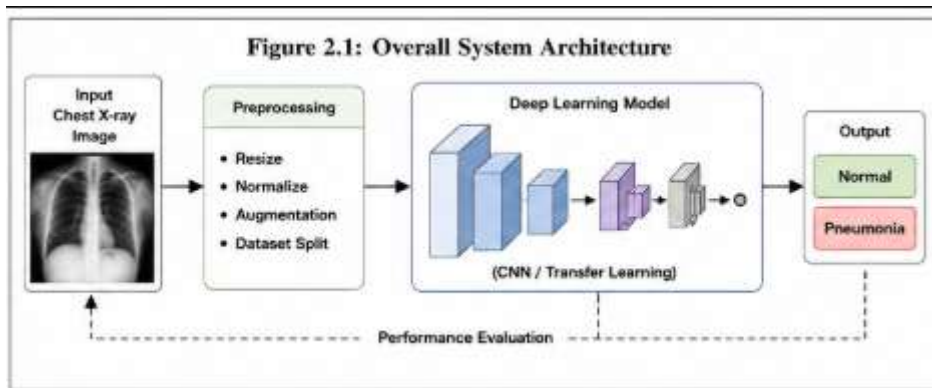
| Module   | Student                    | Main Work                               |
|----------|----------------------------|-----------------------------------------|
| Module 1 | D. Bindu Priya (192425162) | Data Collection and Image Preprocessing |
| Module 2 | K. Gowthami (192525428)    | Deep Learning Model Development         |
| Module 3 | M. Sri Lakshmi (192525404) | Explainable AI and Prediction           |

## CHAPTER 2

### LITERATURE REVIEW AND EXISTING SOLUTIONS

#### 2.1 Review Method

The literature review considers established approaches in chest X-ray analysis, convolutional neural networks, transfer learning, medical image preprocessing and explainable artificial intelligence. The purpose is to understand the strengths and limitations of existing approaches and to identify a practical architecture for an academic prototype.



#### 2.2 Existing Approaches

Traditional computer-aided diagnosis systems often relied on handcrafted features, texture descriptors and conventional classifiers. These approaches can be useful as baselines but may require carefully designed features and may not generalize across acquisition conditions.

Deep learning changed the workflow by allowing image features to be learned jointly with the classifier. CNN-based systems can learn low-level edges and textures in early layers and progressively more abstract patterns in deeper layers. Transfer learning is especially useful when the medical dataset is smaller than the datasets used to train large general-purpose networks.

More recent systems also consider localization and explainability. Rather than returning only a class label, explainable methods can provide heatmaps that indicate image regions associated with the model's output. Such methods can improve transparency, although a heatmap should not be interpreted as definitive evidence of pathology.

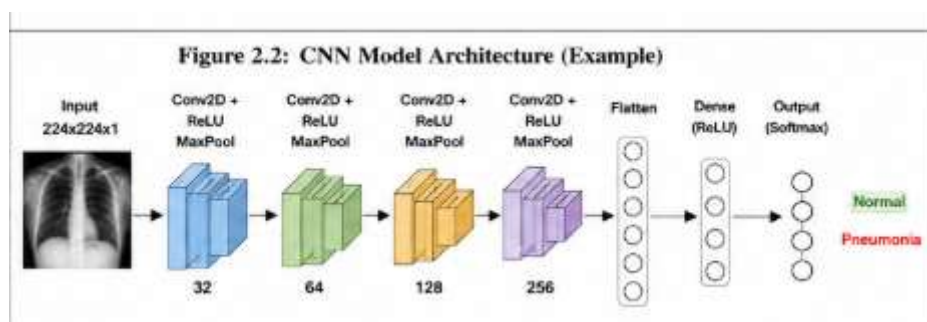
#### 2.3 Comparative Analysis

| Existing Approach                           | Advantages                              | Limitations                                           | Relevance                                        |
|---------------------------------------------|-----------------------------------------|-------------------------------------------------------|--------------------------------------------------|
| Handcrafted features + classical classifier | Simple baseline; low computational cost | Feature engineering dependent; limited representation | Useful for baseline comparison                   |
| CNN from scratch                            | Learns task-specific image features     | Can require more data and compute                     | Suitable when enough training data are available |
| Transfer learning                           | Faster convergence;                     | Pretrained features                                   | Practical choice for                             |

|                        |                                            |                                     |                                      |
|------------------------|--------------------------------------------|-------------------------------------|--------------------------------------|
|                        | effective with limited data                | may introduce domain mismatch       | prototype development                |
| Ensemble models        | Can improve robustness by combining models | Higher compute and complexity       | Possible future enhancement          |
| Explainable AI methods | Provides visual interpretation             | Explanation is not a clinical proof | Important for transparent prediction |

## 2.4 Research / Engineering Gap

A practical gap remains between high-performing research models and systems that are simple, reproducible and explainable enough for student-level deployment. A complete solution must treat the dataset, preprocessing, model, evaluation and explanation as one pipeline. Performance alone is insufficient if the model learns dataset-specific shortcuts or if its predictions cannot be meaningfully inspected.



## 2.5 Proposed Contribution

The proposed project contributes a modular workflow that separates the image-preparation responsibilities of Module 1 from the model-development responsibilities of Module 2 and the explainability and prediction responsibilities of Module 3. This separation makes the system easier to test and modify.

## 2.6 Module 1 Literature Perspective

Data quality is a central factor in medical image classification. Images should be checked for corruption, inconsistent dimensions, duplicated content and inappropriate labels. Preprocessing must preserve medically relevant structures while producing inputs that are compatible with the selected model. Augmentation should be conservative so that generated images remain clinically plausible.

## 2.7 Module 2 Literature Perspective

CNNs are widely used for image classification because their convolutional operations exploit spatial structure. Transfer-learning architectures can provide strong initial representations, after which the final layers are adapted to the target classes. Training should use validation data for model selection and an untouched test set for final evaluation.

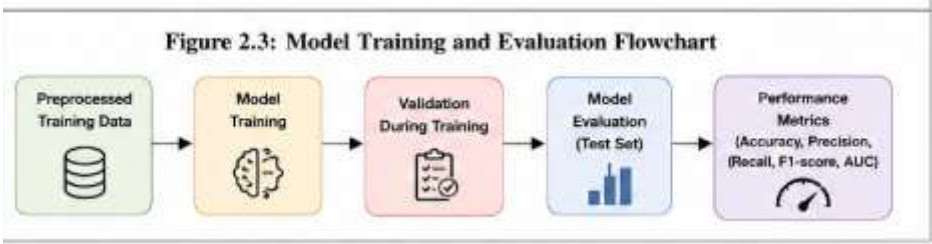
## 2.8 Module 3 Literature Perspective

Explainable AI methods attempt to make model behaviour more understandable. Grad-CAM is a commonly used

approach for CNN-based image models and produces a class-specific localization map based on gradients flowing into a convolutional layer. In this project, the explanation is intended to help users inspect whether the model is focusing on a meaningful lung region rather than irrelevant image artifacts.

### 2.9 Summary of Literature Review

The review supports the use of a carefully prepared image dataset, standardized preprocessing, transfer learning or a compact CNN, and an explainability layer. It also emphasizes the need for independent testing, transparent reporting of limitations and avoidance of unsupported clinical claims.





## CHAPTER 3

### ENGINEERING DESIGN, TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

#### 3.1 Stakeholder / User Requirements

| Stakeholder                    | Requirement                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------|
| Student / developer            | Modular, reproducible and testable software pipeline.                                                     |
| Medical-domain user            | Clear prediction output with supporting visual explanation.                                               |
| Project supervisor / evaluator | Documented methodology, measurable evaluation and traceable contributions.                                |
| Data owner                     | Appropriate handling and acknowledgement of dataset sources.                                              |
| Patient / subject              | Privacy-conscious use of medical images and avoidance of unnecessary exposure of identifying information. |

#### 3.2 Functional and Non-Functional Requirements

| Type           | Requirement                       | Target                                        |
|----------------|-----------------------------------|-----------------------------------------------|
| Functional     | Accept chest X-ray image as input | Supported image format and valid dimensions   |
| Functional     | Preprocess image                  | Consistent resize and normalization           |
| Functional     | Predict class                     | Normal / Pneumonia                            |
| Functional     | Generate explanation              | Grad-CAM or comparable visual map             |
| Non-functional | Reproducibility                   | Fixed preprocessing and documented parameters |
| Non-functional | Usability                         | Simple prediction display                     |
| Non-functional | Safety                            | Clearly state decision-support limitation     |

3.3 Design Specifications

| Parameter           | Target / Requirement                                     | Priority |
|---------------------|----------------------------------------------------------|----------|
| Input image         | Chest X-ray, standardized size                           | High     |
| Classes             | Normal / Pneumonia                                       | High     |
| Image preprocessing | Resize + normalization + quality checks                  | High     |
| Augmentation        | Rotation/flip/brightness only when clinically reasonable | Medium   |
| Model               | CNN or transfer-learning classifier                      | High     |
| Evaluation          | Accuracy, precision, recall, F1, confusion matrix        | High     |
| Explainability      | Class-specific heatmap                                   | High     |
| Deployment          | Notebook / local application prototype                   | Medium   |

3.4 Alternative Solutions & Evaluation

| Criterion                    | Classical ML | CNN from Scratch | Transfer Learning             |
|------------------------------|--------------|------------------|-------------------------------|
| Feature learning             | Manual       | Automatic        | Automatic                     |
| Data requirement             | Moderate     | High             | Moderate                      |
| Training complexity          | Low          | High             | Medium                        |
| General image representation | Limited      | Task-specific    | Strong initial representation |
| Project suitability          | Medium       | Medium           | High                          |

3.5 Selected Design

A CNN-based transfer-learning approach is selected as the preferred design because it offers a practical balance between representation quality, training time and dataset requirements. The final architecture should be chosen after considering available compute, image resolution and validation performance.

Module 1 provides the standardized image tensor. Module 2 receives the prepared data and produces a trained classifier. Module 3 receives the classifier and input image and returns the predicted class, probability information and explanation map.

### 3.6 Preprocessing Design

The preprocessing pipeline includes image loading, format validation, conversion to the required channel representation, resizing to the model input dimensions, intensity normalization, optional contrast handling, and dataset partitioning. Augmentation is applied only to training images so that validation and test sets remain representative of unseen data.

### 3.7 Trade-off Analysis

| Design Decision | Alternative                | Benefit                     | Disadvantage                          | Final Decision                                |
|-----------------|----------------------------|-----------------------------|---------------------------------------|-----------------------------------------------|
| Image size      | Very high resolution       | Retains more detail         | Higher memory and compute cost        | Use model-appropriate standardized size       |
| Model           | Large deep network         | Potentially higher capacity | Slower and more prone to overfitting  | Use transfer learning / compact architecture  |
| Augmentation    | Aggressive transformations | More variation              | May create unrealistic medical images | Use conservative augmentation                 |
| Explanation     | Prediction only            | Simple output               | Poor transparency                     | Prediction + Grad-CAM                         |
| Deployment      | Cloud inference            | More compute                | Privacy/connectivity concerns         | Prototype locally / in controlled environment |

### 3.8 Sustainability

The project can reduce unnecessary computational cost by using transfer learning, image-size standardization and efficient training schedules. Reusable open-source software tools can support reproducibility. Dataset storage should be managed carefully to avoid unnecessary duplication of large image files.

### 3.9 Ethics and Privacy

Medical images can contain sensitive information and must be handled according to the terms of the dataset and applicable institutional requirements. The project should avoid publishing patient-identifying information. If metadata are not required for the experiment, they should not be retained in the prediction interface.

### 3.10 Safety and Clinical Responsibility

A model can produce false positives and false negatives. A false negative may incorrectly classify a pneumonia case as normal, while a false positive may create unnecessary concern or additional review. Therefore, the system must be presented as a decision-support prototype rather than an autonomous diagnostic authority. Clinical decisions must

remain with qualified healthcare professionals.

**3.11 Security**

The software environment should restrict access to datasets and trained models, avoid unnecessary external transmission of images and use controlled storage. If a web interface is developed, uploaded images should be handled securely and temporary files should be deleted when no longer required.

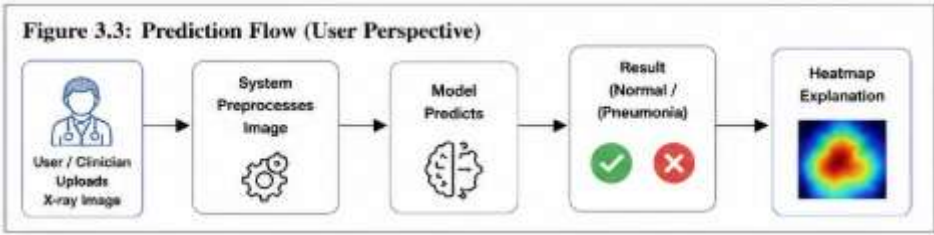
3.12 Risk Register

| Risk                          | Likelihood | Severity | Mitigation                                                          | Residual Risk |
|-------------------------------|------------|----------|---------------------------------------------------------------------|---------------|
| Incorrect / noisy labels      | Medium     | High     | Review dataset documentation and inspect samples                    | Medium        |
| Data leakage between splits   | Low        | High     | Create patient-level or source-aware splits where metadata permit   | Low           |
| Class imbalance               | Medium     | Medium   | Use balanced sampling / class weights and report class-wise metrics | Low-Medium    |
| Poor generalization           | Medium     | High     | Augmentation, regularization and external validation                | Medium        |
| Shortcut learning / artifacts | Medium     | High     | Inspect Grad-CAM maps and test across sources                       | Medium        |
| False negative                | Medium     | High     | Prioritize sensitivity/recall analysis and human review             | Medium        |
| Privacy exposure              | Low        | High     | Local processing and removal of unnecessary metadata                | Low           |

3.13 System Architecture

The proposed architecture follows the flow: Chest X-ray Input → Data Validation → Preprocessing → Augmentation for Training → Deep Learning Classifier → Prediction Probabilities → Explainability Layer → Result Display. The

architecture separates data preparation, model inference and interpretation so that each component can be independently evaluated.



**3.14 Engineering Design Summary**

The selected design emphasizes reproducibility and responsible interpretation. The system is not evaluated only on a single accuracy value; instead, it should be examined through class-wise metrics, confusion matrices, probability outputs and explanation maps. This broader evaluation is important because medical AI performance can be misleading when class imbalance or dataset bias is ignored.

## CHAPTER 4

### IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

#### 4.1 Development Approach & Resources

The system follows an iterative build-and-test methodology. Module 1 first prepares the dataset and verifies the preprocessing pipeline. Module 2 then trains and validates the model using the prepared images. Module 3 integrates the trained model into a prediction and explainability pipeline. Each stage is tested before integration.

| Resource                      | Description                                       | Stage           |
|-------------------------------|---------------------------------------------------|-----------------|
| Chest X-ray dataset           | Normal and pneumonia-labelled radiographic images | Data collection |
| Python                        | Programming and experiment environment            | All modules     |
| OpenCV / PIL                  | Image loading and preprocessing                   | Module 1        |
| TensorFlow / Keras or PyTorch | Deep learning model development                   | Module 2        |
| GPU runtime                   | Accelerated model training                        | Module 2        |
| Matplotlib / similar          | Curves, confusion matrix and visualization        | Modules 2–3     |
| Grad-CAM implementation       | Class-specific explanation heatmaps               | Module 3        |

#### 4.2 Module 1 – Data Collection and Image Preprocessing

D. Bindu Priya is responsible for establishing the input data pipeline. The first step is to collect a dataset containing chest X-ray images with labels appropriate to the project objective. The dataset is then organized into a consistent directory or metadata structure. Each image is checked for readability, inappropriate dimensions, corruption and obvious duplication where feasible.

Images are resized to the model's expected input dimensions. Pixel values are normalized so that the model receives inputs in a stable numerical range. Training augmentation can include small rotations, horizontal flipping where medically appropriate, mild brightness variation and other conservative transformations. Validation and test images are not augmented in the same way because their role is to represent unseen data.

#### 4.3 Dataset Preparation Procedure

- Acquire the permitted chest X-ray dataset and review its documentation.

- Verify labels and identify missing, corrupted or unreadable files.
- Separate training, validation and test data without leakage.
- Resize images to the chosen model input size.
- Normalize pixel intensities consistently.
- Apply augmentation only to the training split.
- Record class counts and preprocessing parameters for reproducibility.



4.4 Module 2 – Deep Learning Model Development

K. Gowthami is responsible for the deep learning development stage. The model receives standardized chest X-ray tensors from Module 1. A CNN or transfer-learning architecture is configured with a final classification layer suitable for the project classes. The training process includes a loss function, optimizer, learning-rate strategy and validation monitoring.

Transfer learning can be implemented by loading a pretrained convolutional backbone, freezing some early layers initially and training a new classification head. Fine-tuning can then be considered after the initial training stage. Regularization techniques such as dropout, weight decay or early stopping can be used when supported by validation evidence.

4.5 Model Training Workflow

| Step | Activity                                                          |
|------|-------------------------------------------------------------------|
| 1    | Load preprocessed training and validation datasets.               |
| 2    | Initialize CNN / transfer-learning architecture.                  |
| 3    | Configure classification head and loss function.                  |
| 4    | Train using mini-batches and validation monitoring.               |
| 5    | Save the best model according to a selected validation criterion. |
| 6    | Evaluate once on the held-out test set.                           |
| 7    | Export the model for Module 3 prediction and explanation.         |

4.6 Hyperparameter Considerations

Important parameters include input image size, batch size, learning rate, number of epochs, optimizer choice, dropout rate and the degree of fine-tuning. These parameters should be selected using validation performance rather than repeatedly tuning against the test set. The final report should record the actual values used in the executed experiment.

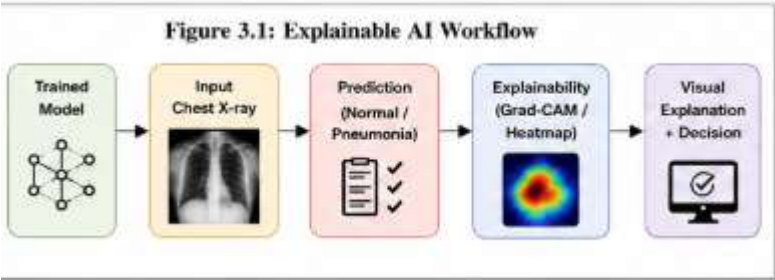
4.7 Evaluation Metrics

Accuracy measures the proportion of correct predictions overall. Precision indicates how many predicted positive cases are actually positive, while recall indicates how many positive cases are successfully detected. F1-score combines precision and recall. The confusion matrix provides class-by-class error counts. ROC-AUC may also be used for binary classification when probability scores are available.

4.8 Module 3 – Explainable AI and Prediction

M. Sri Lakshmi is responsible for the prediction and explainability stage. After the model is trained, a new chest X-ray is passed through the same preprocessing pipeline used during training. The model returns a probability distribution over the target classes. The highest-probability class is displayed together with the corresponding probability, while the explanation layer generates a visual map.

Grad-CAM can be used to produce a class-specific heatmap from a suitable convolutional layer. The heatmap is resized to the original image dimensions and overlaid on the X-ray with an appropriate transparency. The resulting visualization is intended to show which regions influenced the model's decision. It should not be described as a definitive lesion segmentation or a proof of clinical pathology.



4.9 Prediction Workflow

| Stage          | Output                                         |
|----------------|------------------------------------------------|
| Input          | Chest X-ray image                              |
| Preprocessing  | Standardized image tensor                      |
| Inference      | Class probabilities                            |
| Decision       | Predicted class                                |
| Explainability | Class-specific heatmap                         |
| Display        | Image + prediction + probability + explanation |

4.10 Result Interpretation

A robust result should be interpreted together with the evaluation dataset and not from an individual prediction alone. If the model predicts pneumonia with high probability, the output can be flagged for review, but the final clinical interpretation should consider patient history, symptoms, examination and professional radiological assessment.

4.11 Testing and Verification

| Requirement                | Test Method                                 | Acceptance Criterion                      |
|----------------------------|---------------------------------------------|-------------------------------------------|
| Preprocessing consistency  | Run identical sample through pipeline twice | Same standardized output                  |
| Classification performance | Evaluate on held-out test set               | Report accuracy, precision, recall and F1 |
| Class-wise behaviour       | Inspect confusion matrix                    | No class hidden by aggregate              |

|                 |                                             |                                           |
|-----------------|---------------------------------------------|-------------------------------------------|
|                 |                                             | accuracy                                  |
| Explainability  | Generate heatmap for representative samples | Heatmap generated without runtime failure |
| Inference       | Run unseen X-ray through complete pipeline  | Prediction displayed correctly            |
| Reproducibility | Repeat using documented settings            | Comparable evaluation result              |

#### 4.12 Results and Analysis

The final numerical results must be populated from the executed experiment rather than estimated in advance. The following table is provided as the reporting structure for the actual project results.

| Metric               | Required Reporting                    |
|----------------------|---------------------------------------|
| Test accuracy        | [Enter measured value]                |
| Precision            | [Enter measured value]                |
| Recall / Sensitivity | [Enter measured value]                |
| F1-score             | [Enter measured value]                |
| ROC-AUC              | [Enter measured value, if calculated] |
| False-positive rate  | [Enter measured value]                |
| False-negative rate  | [Enter measured value]                |
| Inference time       | [Enter measured value, if measured]   |

#### 4.13 Confusion Matrix Analysis

The confusion matrix should be included after the final model has been evaluated. For the binary Normal/Pneumonia task, the matrix should show true negatives, false positives, false negatives and true positives. Special attention should be given to false negatives because failing to flag a pneumonia-related image may be more consequential than a false positive in a screening-support scenario.

#### 4.14 Explainability Analysis

Representative Grad-CAM outputs should be examined to determine whether the model is focusing on plausible lung regions. If strong activation consistently appears on image borders, text markers, acquisition artifacts or other irrelevant regions, the model may be learning shortcuts. Such observations should be documented as limitations and used to guide further dataset and preprocessing improvements.

#### 4.15 Project Planning, Roles and Timeline

| Phase                | Indicative Period | Responsible Module |
|----------------------|-------------------|--------------------|
| Requirement analysis | Weeks 1–2         | All members        |

|                               |             |             |
|-------------------------------|-------------|-------------|
| Literature review             | Weeks 2–3   | All members |
| Dataset preparation           | Weeks 3–5   | Module 1    |
| Model design and training     | Weeks 5–8   | Module 2    |
| Explainability and prediction | Weeks 8–10  | Module 3    |
| Testing and evaluation        | Weeks 10–12 | All members |
| Report compilation            | Weeks 12–13 | All members |

#### 4.16 Achievement of Project Objectives

| Objective                         | Evidence to be Presented                                       | Status             |
|-----------------------------------|----------------------------------------------------------------|--------------------|
| Collect and preprocess X-ray data | Dataset structure, preprocessing parameters and sample outputs | To be verified     |
| Develop deep learning classifier  | Training configuration and saved model                         | To be verified     |
| Evaluate model                    | Held-out test metrics and confusion matrix                     | To be verified     |
| Generate explainable prediction   | Prediction screen and Grad-CAM examples                        | To be verified     |
| Document responsible use          | Ethics, safety and limitation discussion                       | Achieved in report |

#### 4.17 Strengths

- Modular structure allows independent testing of data, model and explanation stages.
- Deep learning can learn complex image representations without manually defining every feature.
- Explainability adds a visual interpretation layer to the prediction.
- The workflow is suitable for academic experimentation and can be extended to other chest X-ray classification tasks.

#### 4.18 Limitations

- Dataset bias may reduce performance when the model is used on images from another hospital or device.
- Image-level labels do not necessarily provide precise lesion locations.
- A high test accuracy alone does not establish clinical usefulness.
- Explainability heatmaps can be approximate and should not be treated as ground-truth pathology.
- External clinical validation and prospective testing are required before real-world medical deployment.

#### 4.19 Project Cost / Resource Considerations

The academic prototype can be developed largely with open-source software and notebook-based GPU resources. Actual costs depend on the selected compute environment and whether a local GPU, cloud notebook or institutional infrastructure is used. Any purchased hardware or paid compute resource should be recorded in the final project budget.

## CHAPTER 5

### CONCLUSION, FUTURE WORK AND REFLECTION

#### 5.1 Conclusion

This project presents an intelligent pipeline for pneumonia detection and diagnosis support from chest X-ray images using deep learning. The system is organized into three complementary modules: data collection and image preprocessing by D. Bindu Priya, deep learning model development by K. Gowthami, and explainable AI with prediction by M. Sri Lakshmi.

The proposed workflow demonstrates how a medical image can move from raw data through quality checking, preprocessing and augmentation to model inference and interpretable prediction. The design recognizes that medical AI requires more than a classifier: data quality, independent evaluation, error analysis, privacy, safety and transparency are equally important.

The final numerical performance should be reported only after the complete experiment has been executed. The evaluation should include class-wise metrics and a confusion matrix, while explainability results should be reviewed for evidence of meaningful attention. The system should remain a decision-support prototype rather than an autonomous diagnostic system.

#### 5.2 Future Work

- Validate the model on independent datasets from different institutions and imaging devices.
- Expand the task from binary pneumonia classification to multiple thoracic conditions.
- Investigate patient-level data splitting and stronger leakage prevention.
- Explore calibration methods so that predicted probabilities better reflect uncertainty.
- Combine image analysis with relevant clinical information where ethically and legally appropriate.
- Improve explainability using multiple complementary methods and compare explanations against expert annotations.
- Optimize the model for efficient inference while maintaining clinically relevant performance.
- Conduct prospective clinical evaluation before considering real-world deployment.

### 5.3 Professional Learning & Individual Reflection

#### D. Bindu Priya (192425162) – Module 1

Working on data collection and image preprocessing develops an understanding that model performance begins with data quality. The module involves organizing image files, checking labels, standardizing dimensions and ensuring that the same preprocessing logic is used during training and prediction. A major learning outcome is recognizing that augmentation should increase useful variation without creating medically unrealistic images. The work also highlights the importance of preventing data leakage and documenting every preprocessing decision for reproducibility.



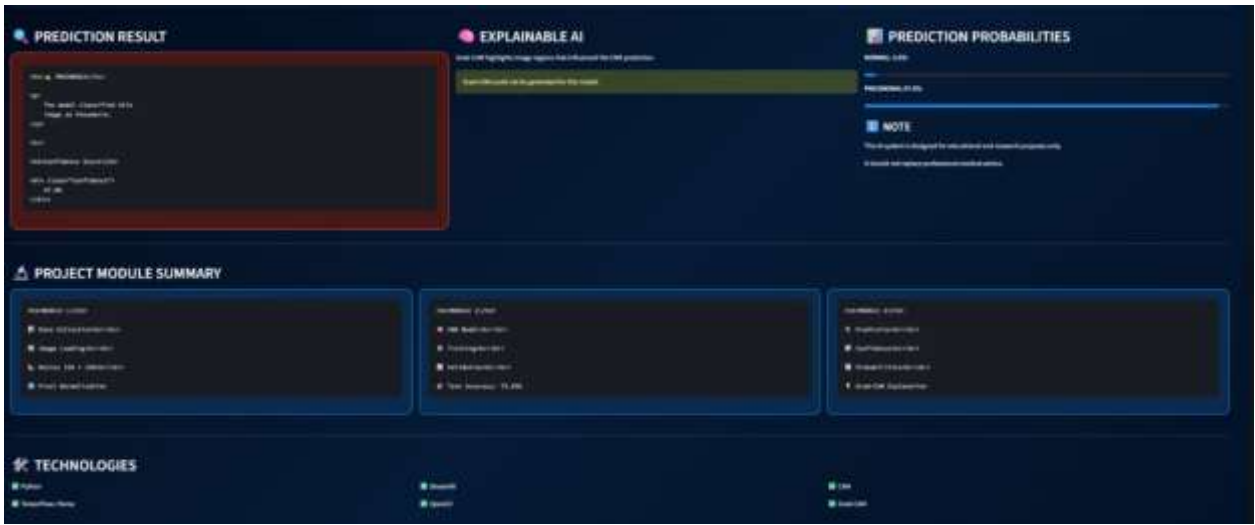
#### K. Gowthami (192525428) – Module 2

The model-development module provides practical experience in CNN architecture, transfer learning, training configuration and evaluation. An important engineering lesson is that a more complex model is not automatically a better model. Validation performance, computational resources, overfitting and generalization must be considered together. The module also develops skills in interpreting training curves and confusion matrices rather than depending on a single accuracy number.



#### M. Sri Lakshmi (192525404) – Module 3

The explainable-AI and prediction module demonstrates that an AI system should communicate not only what it predicts but also provide a meaningful way to inspect the prediction. Implementing probability output and Grad-CAM develops practical understanding of model interpretation. The module also emphasizes responsible communication: an explanation heatmap is an aid for inspection and should not be presented as a medical diagnosis or exact lesion boundary.



5.4 Team Reflection

The project demonstrates the value of dividing a complex engineering problem into connected modules. Data preparation, model development and explainability depend on one another, so interface definitions and shared documentation are important. The team should maintain consistent preprocessing parameters, model versions and evaluation procedures so that results remain traceable.



## REFERENCES

- [1] R. Siddiqi and S. Javaid, “Deep Learning for Pneumonia Detection in Chest X-ray Images: A Comprehensive Survey,” *Journal of Imaging*, vol. 10, no. 8, p. 176, 2024. doi: 10.3390/jimaging10080176.
- [2] G. Luo, Z. Wang, Y. Wang, and K. Wang, “Detection of pneumonia in chest X-ray image based on separable convolutional neural network,” *Concurrency and Computation: Practice and Experience*, vol. 36, no. 18, e8160, 2024. doi: 10.1002/cpe.8160.
- [3] Q. An, W. Chen, W. Shao, et al., “A Deep Convolutional Neural Network for Pneumonia Detection in X-ray Images with Attention Ensemble,” *Diagnostics*, vol. 14, no. 4, p. 390, 2024. doi: 10.3390/diagnostics14040390.
- [4] R. Vyas and D. R. Khadatkhar, “Ensemble of Deep Learning Architectures with Machine Learning for Pneumonia Classification Using Chest X-rays,” *Journal of Digital Imaging*, vol. 38, no. 2, pp. 727–746, 2024. doi: 10.1007/s10278-024-01201-y.
- [5] J. Hofmeister and N. Garin, et al., “Validating the accuracy of deep learning for the diagnosis of pneumonia on chest x-ray against a robust multimodal reference diagnosis: a post hoc analysis of two prospective studies,” *European Radiology Experimental*, vol. 8, article 20, 2024. doi: 10.1186/s41747-023-00416-y.
- [6] M. Fu, C. Tantithamthavorn, and T. Le, “DAViT: A Domain-Adapted Vision Transformer for Automated Pneumonia Detection and Explanation Using Chest X-Ray Images,” *IEEE Access*, vol. 13, pp. 103033–103044, 2025. doi: 10.1109/ACCESS.2025.3579314.
- [7] J. Colin and N. Surantha, “Interpretable Deep Learning for Pneumonia Detection Using Chest X-Ray Images,” *Information*, vol. 16, no. 1, p. 53, 2025. doi: 10.3390/info16010053.
- [8] S. Biswas, R. Mostafiz, and M. S. Uddin, et al., “FLPneXAINet: Federated deep learning and explainable AI for improved pneumonia prediction utilizing GAN-augmented chest X-ray data,” *PLOS ONE*, vol. 20, no. 7, e0324957, 2025. doi: 10.1371/journal.pone.0324957.
- [9] E. Yazdani, A. Neizehbaz, N. Karamzade-Ziarati, and S. R. Kheradpisheh, “Explainable artificial intelligence for pneumonia classification: Clinical insights into deformable prototypical part network in pediatric chest x-ray images,” *Journal of Medical Imaging and Radiation Sciences*, vol. 56, no. 5, p. 102023, 2025. doi: 10.1016/j.jmir.2025.102023.
- [10] P. Chedsom, “Pneumonia Detection from Chest X-Ray Images using Convolutional Neural Networks and Transfer Learning Techniques,” *Journal of Applied Informatics and Technology*, 2025. doi: 10.14456/jait.2025.25.

- [11] D. Saranyaraj, V. Shrinaath, A. Nayak, et al., “PneuNet: a lightweight convolutional neural network with multiscale feature fusion for automated pneumonia detection from chest X-rays,” *Frontiers in Medicine*, vol. 12, 1713587, 2026. doi: 10.3389/fmed.2025.1713587.
- [12] Z. Xu and H. Zhang, “Pneumonia detection from enhanced chest X-Ray images based on Double SGAN model,” *Scientific Reports*, vol. 16, article 9922, 2026. doi: 10.1038/s41598-026-39785-w.
- [13] G. Deshmuk, V. Kulkarni, V. U. Rathod, A. Chhajer, A. Pawar, and V. Ghonge, “Optimized deep learning approach for pneumonia detection using chest X-ray images,” *Nigerian Journal of Technology*, vol. 45, no. 1, pp. 1–13, 2026. doi: 10.4314/njt.v45i1.1s.
- [14] B. Mahtabi, E. Nasr-Esfahani, and S. Yaraghi, “Parameter-efficient deep learning for pneumonia detection on chest X-rays: A comparative evaluation of explainable AI methods,” *medRxiv*, 2026. doi: 10.64898/2026.07.14.26358065.
- [15] “An intelligent decision support framework for automated pneumonia classification from chest X-ray images using hybrid vision transformers and CNNs,” *Intelligence-Based Medicine*, vol. 14, 100385, 2026.

## **APPENDICES**

### **Appendix A – Detailed Preprocessing Procedure**

Document the final input dimensions, normalization range, augmentation parameters, dataset split procedure and class counts used in the executed experiment. Include representative before-and-after images if required.

### **Appendix B – Model Configuration**

Record the selected CNN or transfer-learning architecture, pretrained weights, trainable/frozen layers, optimizer, learning rate, batch size, number of epochs, loss function, callbacks and model checkpoint criteria.

### **Appendix C – Prediction and Explainability**

Include the final prediction interface or notebook output, class probabilities and representative Grad-CAM visualizations. Explain how the heatmap was generated and which convolutional layer was used.

### **Appendix D – Source Code Information**

Include the essential code excerpts for image preprocessing, dataset loading, model training, evaluation, prediction and Grad-CAM. The complete source code should be maintained in the project's version-controlled repository or submitted according to departmental requirements.

### **Appendix E – Experimental / Raw Data**

Attach the raw evaluation metrics, confusion matrix values, test-set summary and relevant screenshots generated during model testing.

## CAPSTONE PROJECT OUTCOME MAPPING SHEET

| Outcome Evidence                                   | SO / Area   | Location in Report             |
|----------------------------------------------------|-------------|--------------------------------|
| Complex engineering problem formulation            | SO1         | Ch. 1, Ch. 2                   |
| Engineering design under constraints               | SO2         | Ch. 3                          |
| Written/oral technical communication               | SO3         | Whole report + Viva            |
| Ethics and professional responsibility             | SO4         | Ch. 3                          |
| Teamwork and leadership                            | SO5         | Ch. 4                          |
| Experimentation, analysis and engineering judgment | SO6         | Ch. 4                          |
| Independent acquisition of new knowledge           | SO7         | Ch. 5                          |
| Standards / responsible data handling              | SO2/SO4     | Ch. 3                          |
| Sustainability and societal impact                 | SO2/SO4     | Ch. 3                          |
| Risk assessment and mitigation                     | SO2/SO4     | Ch. 3                          |
| Security considerations                            | Relevant SO | Ch. 3                          |
| Integrated/system-level engineering                | SO1/SO2     | Ch. 3                          |
| Project planning and management                    | SO5         | Ch. 4                          |
| Individual contribution                            | SO1–SO7     | Contribution Statement + Ch. 4 |